Apr. 21

- Groups defined
  - Classified cyclic groups – 1 subgroup per divisor of order
  - Classified Abelian groups

    Ex. Abelian group G of order 18

    $G \cong \mathbb{Z}/18\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/9\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$
  - Quotient groups: use a normal subgroup to simplify a group
  - Want to classify groups w/o normal subgroups: Simple groups
    - Classification of all finite simple groups ex. A_n always simple
  - Sporadic groups don't fit into any of the classified families of groups
    - "Monster group" largest simple group

- Integral domains are rings more like $\mathbb{Z}$
  - What about rings that are more like $\mathbb{Q}$? fields

- <u>Def</u> A <u>field</u> is a commutative ring with unity ($1 \neq 0$, must be nonzero for the zero ring to not be a field, want + identity and · identity to be distinct) in which every nonzero element is a unit

  note: $\mathbb{Z}/2\mathbb{Z}$ is the smallest field, zero ring is the smallest ring

- <u>Alternative def</u> A <u>field</u> F is a commutative ring such that $(F - \{0\}, \cdot)$ is an Abelian group

Ex. $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}/2\mathbb{Z} \rightarrow$ check $\mathbb{Z}/2\mathbb{Z} - \{\bar{0}\} = \{\bar{1}\}$

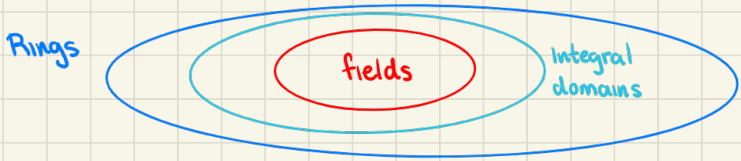
$\mathbb{Q}(\alpha)$ where α is a root of irreducible polynomial in $\mathbb{Q}[x]$

$\subset$ subring of $\mathbb{C}$, eg $\mathbb{Q}(\sqrt{2}) = \{r + s\sqrt{2} : r, s \in \mathbb{Q}\}$

- In general, $\mathbb{Z}/p\mathbb{Z}$ is a field for every prime p

- <u>Prop.</u> Every field is an integral domain

  <u>Proof</u> Suppose F is a field and let $a, b \in F$ with $a \neq 0$ and $ab = 0$. Want to show b must be 0. Since F is a field and $a \neq 0$, $a^{-1} \in F$. We see $a^{-1}ab = a^{-1} \cdot 0$, so $b = 0$. $\square$



- **Thm 13.2** Every finite integral domain is a field

  Want to show every nonzero element is a unit

  Proof Let D be a finite integral domain and $a \in D$ be nonzero.
  We see $a^n \in D$ for every integer $n > 0$. Since D is finite, $a^m = a^k$ for some $m > k > 0$. We have $a \cdot a^{m-1} = a \cdot a^{k-1}$ where if $k-1=0$ then a^{k-1} divides 1.
  By cancellation, we have $a^{m-1} = a^{k-1}$. We continue in this manner until $a^{m-k+1} = a^{k-k+1} = a$.
  Thus $a \cdot a^{m-k} = a \cdot 1$, so $a^{m-k} = 1$. Therefore $a^{m-k-1} \cdot a = 1$, where if $m-k-1=0$, then $a^0 = 1$. Thus $a^{m-k-1} = a^{-1}$.
  We have shown an arbitrary nonzero element of D $\underbrace{\text{has a multiplicative inverse}}_{\text{is a unit}}$, so D is a field. $\square$

Ex. Suppose $a^5 = a^3$

Then cancel an a to get $a^4 = a^2$, then $a^3 = a$, then $a^2 = 1 \Rightarrow a \cdot a = 1 \Rightarrow a^{-1} = a$

- Infinite integral domains that aren't fields $\mathbb{Z}$, $\mathbb{Z}[i] = \{a+bi : a,b \in \mathbb{Z}\}$, $\mathbb{Z}[\sqrt{2}]$, etc

- **Corollary** $\mathbb{Z}/p\mathbb{Z}$ is a field for every prime p We often denote this field $\mathbb{F}_p$

  To prove: $\mathbb{Z}/p\mathbb{Z}$ is an integral domain

- fact: every finite field has a prime power number of elements

Ex. $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}\}$, $\mathbb{F}_3(i) = \{a+bi : a,b \in \mathbb{F}_3 \text{ and } i^2 = -\bar{1} = \bar{2}\}$

$\mathbb{F}_3(i)$ has 9 elements, arithmetic below done in $\mathbb{Z}/3\mathbb{Z}$

Addition: $(1+2i)+(0+2i) = 1+0+(2+2)i = 1+1i = 1+i$

Mult.: $(1+2i)(2+2i) = 2+4i+2i+4i^2 = 2+6i+4(-1) = 2-4+6i = -2+0i = 1$

- Aside: inverses in $\mathbb{C}$

  Suppose a,b not both 0

  $\underbrace{(a+bi)(a-bi)}_{a^2+b^2} = a^2 + abi - abi - b^2 i^2 = \frac{a^2+b^2}{a^2+b^2}$ so $(a+bi)^{-1} = \frac{a-bi}{a^2+b^2}$

- In $\mathbb{F}_3(i)$: $i+i+i = 3i = 0$, strange things can happen even in fields

- **Def** the <u>characteristic</u> of a ring R is the least positive integer n such that $\underbrace{a+a+\ldots+a}_{n \text{ times}} = 0$ $\forall a \in R$
  - If no such integer exists, we say the ring has characteristic 0
  - Textbook uses $n \cdot a$, we like $[n]a$ for $\underbrace{a+a+\ldots+a}_{n \text{ times}} = 0$
  - Often abbreviate as char R

Ex. $\mathbb{Z}/n\mathbb{Z}$ has characteristic n, char $\mathbb{Z} = 0$, char $\mathbb{F}_3(i) = 3$

- **Thm. 13.3** Let R be a ring with unity 1. If $|1|$ in $(R,+)$ is ∞, then $\operatorname{char} R = 0$. If $|1| = n < \infty$, then $\operatorname{char} R = n$.

  order

  1 tells you everything you need to know about char R

  Proof sketch

  $|1| = \infty$ case follows from def. of char

  $|1|$ finite case: Let $a \in R$. Consider $\underbrace{a + a + \dots + a}_{n \text{ times}} = a(\underbrace{1 + 1 + \dots + 1}_{n \text{ times}}) = a \cdot 0 = 0$.

- **Thm. 13.4** The characteristic of an integral domain is either 0 or prime.

  Proof It suffices to consider the additive order of 1 by the previous theorem.

  If 1 has infinite additive order, then $\operatorname{char} D = 0$ and we are done.

  Suppose 1 has finite additive order n. By Thm. 13.3, D has characteristic n.

  Suppose $n = t \cdot s$ with $1 < t, s < n$ (n is not prime). Then $0 = \underbrace{1 + \dots + 1}_{n \text{ times}} = \underbrace{\underbrace{1 + \dots + 1}_{s \text{ times}} + \underbrace{1 + \dots + 1}_{s \text{ times}} + \dots + \underbrace{1 + \dots + 1}_{s \text{ times}}}_{t \text{ times}}$

  $= 1 \cdot s + \dots + 1 \cdot s = (\overbrace{1 + \dots + 1}^{t \text{ times}}) \cdot s = t \cdot s = 0$, which is a contradiction since $t \neq 0, s \neq 0$, D is an integral domain. $\square$